A strict entropy gap gives a bounded causal simulation cost

Research note for the finite-Shannon-theorem project

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Abstract

For a finite stationary family of finite output laws with a unique entropy-maximizing action, we construct exact causal seeds of entropy $nh_{\max} + O(1)$, uniformly in the horizon. The construction works for every deterministic adaptive policy with a common decoder. Moreover, $C_n - nh_{\max}$ increases to a finite limit, which is zero exactly when every other output law is a deterministic image of the maximizing law. The proof combines the project's finite-seed tail bound with a block absorption lemma and an anti-diagonal arrangement. All constants below are explicit. This is a mathematical proof, not a Lean certificate; historical priority is not asserted.

1 Statement and the existing input

Let $P_0, \dots, P_d$ be probability laws on finite nonempty alphabets. Work on their positive supports. At each step an action i requests an independent response with law P_i, conditionally on the observed history. A seed fixes one deterministic response strategy on all histories. Let C_n be the infimum of its Shannon entropy among exact horizon-n seeds. All information quantities use bits; ln and e^x are natural.

Theorem 1. *Suppose $H(P_0) = h > \max_{1 \le i \le d} H(P_i) = h_-$, with $d \ge 1$. There is an explicit finite constant K, given in Section 5, such that for every integer $n \ge 0$,*

$$nh \le C_n \le nh + K.$$

In fact, for some $\ell \in [0, K]$,

$$\boxed{C_n = nh + \ell + o(1), \qquad C_n - nh \uparrow \ell.}$$

Furthermore $\ell = 0$ if and only if, for each $i \ge 1$, there is a map $f_i : \operatorname{supp} P_0 \to \operatorname{supp} P_i$ with $(f_i)_{\#} P_0 = P_i$. In that case $C_n = nh$ for every n. With just one action, the same equality is immediate.

The sole non-elementary input is the project's already proved finite causal-seed estimate. For every positive finite controlled tree T, there is a finite exact seed B whose information $J = -\log_2 \mathbb{P}(B)$ satisfies

$$\mathbb{P}(J > t + s) \le 1 - F_T^*(t) + 2^{-s} \qquad (t \in \mathbb{R},\ s \ge 0). \tag{1}$$

Here F_T^* is the pointwise infimum of the transcript-information CDFs over deterministic adaptive policies. This is `CausalSpectrum.causal_truncation` in `CausalSeed/Entropy.lean`. We use the repository at commit `1889d3b333a97b5ebd7c73ecabfc60826d6afb41`. No equal-entropy hypothesis, asymptotic diffusion estimate, or unspecified FGL constant is needed.

2 A concentrated seed for all submaximal actions

Write $I_i(y) = -\log_2 P_i(y)$ and set

$$M_0 = \max\{1, \max_y |I_0(y) - h|\}, \qquad M_- = \max\{1, \max_{i\geq 1, y} |I_i(y) - h_-|\}.$$

For every $q \geq 1$, choose the same finite seed B_q supplied by (1) for q uses of the subfamily $P_1, \ldots, P_d$. It simulates all adaptive policies of that subfamily, not just a fixed list of its actions. Put $J_q = -\log_2 \mathbb{P}(B_q)$.

For any such policy, each centered information increment belongs to $[-M_-, M_-]$ and has conditional mean at most zero. The conditional Hoeffding bound gives

$$\mathbb{E}e^{\theta \sum_{t=1}^{q} (I_{A_t}(Y_t) - h_-)} \leq e^{q\theta^2 M_-^2/2} \quad (\theta \geq 0).$$

Indeed, condition on the past, apply Hoeffding's lemma on the interval $[-M_-, M_-]$, and iterate. Markov's inequality, optimized at $\theta = v/(qM_-^2)$, therefore gives, uniformly over policies,

$$1 - F_q^*(qh_- + v) \leq e^{-v^2/(2qM_-^2)} \qquad (v \geq 0). \tag{2}$$

Combining this with (1), splitting an excess into two equal parts, yields

$$\mathbb{P}(J_q > q(h_- + \delta) + u) \leq e^{-(q\delta+u)^2/(8qM_-^2)} + 2^{-(q\delta+u)/2} \qquad (\delta > 0,\ u \geq 0). \tag{3}$$

Set $b = h_- + \delta$. Integration of the tail, using $\mathbb{E}[J\mathbf{1}_{J>v}] = v\mathbb{P}(J > v) + \int_v^\infty \mathbb{P}(J > t)\, dt$, gives

$$\begin{aligned} T_q := \mathbb{E}[J_q \mathbf{1}_{J_q > qb}] \leq &\left(qb + \frac{4M_-^2}{\delta}\right) e^{-q\delta^2/(8M_-^2)} \\ &+ \left(qb + \frac{2}{\ln 2}\right) e^{-q\delta \ln 2/2}. \end{aligned} \tag{4}$$

For the first integral use $(q\delta + u)^2 \geq q^2\delta^2 + 2q\delta u$. This tail-entropy bound avoids any dependence on the possibly enormous number of seed atoms.

3 Absorbing a whole adaptive block

Choose, henceforth,

$$\delta = \frac{h - h_-}{4}, \qquad b = h_- + \delta, \qquad \Lambda = \frac{h - \delta}{h_- + 2\delta} > 1. \tag{5}$$

Lemma 2 (Exact block absorption)**.** *For all positive integers p, q with $p \geq q/\Lambda$, there is a joint law of $A \sim P_0^{\otimes p}$ and the exact subfamily seed B_q such that*

$$H(A, B_q) \leq ph + D(q+1)e^{-\gamma q}, \tag{6}$$

where the following constants are strictly positive and explicit:

$$\begin{aligned} \beta &= \min\left\{\frac{\delta^2}{2\Lambda M_0^2}, \delta \ln 2\right\}, \\ \gamma &= \min\left\{\frac{\beta}{2}, \frac{\delta^2}{8M_-^2}, \frac{\delta \ln 2}{2}\right\}, \\ D &= 4b + \frac{2\sqrt{2}}{\ln 2} + \frac{4M_-^2}{\delta} + \frac{2}{\ln 2}. \end{aligned}$$

Proof. Retain the atoms of B_q of mass at least 2^{-qb}; there are at most 2^{qb} of them. Collapse all remaining atoms to one symbol $*$. Call the resulting random variable B'_q and the number of retained atoms k.

Couple A and B'_q by subdividing $[0,1)$ into consecutive intervals of their respective atom masses and using the same uniform point. An A-interval wholly inside one B'_q-interval permits exact decoding of B'_q from A. At most k internal boundaries can split the other A-intervals. Outside the event $I(A) < p(h-\delta)$, each A-interval has length at most $2^{-p(h-\delta)}$. Thus a deterministic decoder has error probability

$$\begin{aligned} e_q &\le e^{-p\delta^2/(2M_0^2)} + k\,2^{-p(h-\delta)} \\ &\le e^{-q\delta^2/(2\Lambda M_0^2)} + 2^{-q\delta} \le 2e^{-\beta q}. \end{aligned}$$

We used Hoeffding's inequality for the iid P_0 information and $p(h-\delta) \ge q(h_- + 2\delta) = qb + q\delta$.

If $k = 0$, the coarse variable is constant. Otherwise its alphabet has at most $k+1$ symbols, so the elementary decoding entropy inequality gives

$$H(B'_q \mid A) \le h_2(e_q) + e_q \log_2 k \le \frac{2\sqrt{2}}{\ln 2} e^{-\beta q/2} + 2qb\, e^{-\beta q}.$$

Here $h_2(u) \le 2\sqrt{u}/\ln 2$ for $0 \le u \le 1$: use $-u\ln u \le (2/e)\sqrt{u}$ and $-(1-u)\ln(1-u) \le u$. These bounds remain valid when the displayed upper bound on e_q exceeds one; no small-error assumption is made.

Refine $*$ by its original conditional distribution, independently of A given B'_q. This preserves both exact marginals. If its mass is v,

$$H(B_q \mid B'_q) = T_q + v \log_2 v \le T_q,$$

with $0 \log 0 = 0$. Consequently $H(B_q \mid A) \le H(B'_q \mid A) + T_q$. Insert (4) and the definitions of D, γ to obtain (6). The case $k = 0$ obeys the same final bound. □

4 Why the couplings remain causal

Let $s_1, \dots, s_m$ be positive integers summing to n, and put $T_j = s_1 + \cdots + s_j$. Draw the following components independently:

$$X^{(1)} \sim P_0^{\otimes s_1}, \quad B_{s_m}, \quad (B_{s_j}, X^{(j+1)}) \quad (1 \le j < m),$$

where the pairs have the exact couplings from Lemma 2. The occurrences of action 0 consume the stream $X^{(1)}, \dots, X^{(m)}$ in forward order. All other occurrences consume the subfamily decoders in block order $m, m-1, \dots, 1$. Within each such block the decoder runs forward, using the local history of subfamily actions and outputs; intervening action-0 calls do not advance it. This defines one deterministic global decoder for each seed value.

Lemma 3 (Separation and accounting)**.** *The decoder is exact for every deterministic adaptive policy. Its entropy is*

$$H(S) \le nh + H(B_{s_m}) + \sum_{j<m} D(s_j+1) e^{-\gamma s_j}, \tag{7}$$

provided $s_{j+1} \ge s_j/\Lambda$.

Proof. Reaching any coordinate of $X^{(j+1)}$ requires at least $T_j + 1$ action-0 calls. Reaching the subfamily block j requires at least $n - T_j + 1$ subfamily calls. Observing both sides of a paired component would therefore require at least $n+2$ calls, which is impossible.

Fix a complete transcript branch of any deterministic policy. The action word on that branch is fixed. The branch event is an intersection of events on independent seed components, and

uses at most one marginal of each pair. A used P_0 marginal gives the product probabilities of its observed prefix. A used subfamily marginal gives the product probabilities of its local action/output path, by exactness of B_{s_j}. A shorter path can be extended to a full local policy and its remaining outputs summed out. Hence the probability of the global branch is exactly the product of the requested transition probabilities. This proves full adaptive exactness, not merely the laws of nonadaptive action words.

Independence of the components and Lemma 2 give

$$H(S) = s_1 h + H(B_{s_m}) + \sum_{j<m} H(B_{s_j}, X^{(j+1)}),$$

which is (7). □

5 An explicit all-horizon bound

Take

$$L = \max\left\{2, \left\lceil \frac{4}{\Lambda - 1} \right\rceil \right\}, \qquad m_- = \max_{i\geq 1, y} I_i(y), \qquad r = e^{-\gamma}.$$

For $n \geq L^2$, choose the largest integer $k \geq L$ such that

$$N_k = k(k-1) - L(L-1) + Lk \leq n.$$

Use the two tails $(L, L+1, \ldots, k-1)$ and its reverse. The remaining length $R = n - [k(k-1) - L(L-1)]$ satisfies

$$Lk \leq R < (L+2)k + L.$$

Set $t = \lfloor R/k \rfloor$, and partition R as evenly as possible into t central blocks. Then $L \leq t \leq L+2$, and each central size s obeys

$$k \leq s \leq k + \left\lceil \frac{k-1}{t} \right\rceil \leq k(1 + 2/L).$$

All adjacent ratios lie in $[1/\Lambda, \Lambda]$. On a tail the ratio is at most $1 + 1/L$; among central blocks it is at most $1 + 2/L$; at a tail/center junction it is at most $(1 + 1/L)(1 + 2/L) \leq 1 + 4/L \leq \Lambda$. The last size is L if there is a tail, and at most $L + 2$ otherwise.

Since every subfamily transcript of length s has information at most sm_-, (1) gives

$$H(B_s) \leq sm_- + \int_0^\infty 2^{-u}\, du = sm_- + \frac{1}{\ln 2}.$$

Every tail length occurs at most twice; there are at most $L + 2$ central blocks. Thus (7) is uniformly bounded by $nh + K_1$, where

$$K_1 = (L+2)m_- + \frac{1}{\ln 2} + (L+4)D\,\frac{r^L[(L+1) - Lr]}{(1-r)^2}. \tag{8}$$

Here we used the exact sum $\sum_{q=L}^\infty (q+1)r^q = r^L[(L+1) - Lr]/(1-r)^2$. For $n < L^2$, independently sample the outputs of every action at each time. This is an exact causal seed of entropy $n\sum_{i=0}^d H(P_i)$. Hence one explicit choice valid for all n is

$$\boxed{K = \max\left\{L^2 \sum_{i=1}^d H(P_i),\ K_1\right\}.} \tag{9}$$

The constant-action-0 policy gives $C_n \geq nh$. This proves the uniform bounds in Theorem 1.

6 The limiting constant and its zero set

Given any exact horizon-$(n+1)$ seed S, request action 0 first and denote its output by Y. Conditional on every $Y = y$ of positive probability, the remaining seed and decoder form an exact horizon-n simulation of the same stationary kernel. Since Y is a function of S,

$$H(S) = h + \sum_y P_0(y) H(S \mid Y = y) \geq h + C_n.$$

Taking infima gives $C_{n+1} \geq h + C_n$. Consequently the residual is nonnegative, nondecreasing and bounded by (9); its finite limit ℓ exists.

If each $P_i = (f_i)_{\#} P_0$, independent P_0 samples, one per time step, give the exact entropy nh by applying the requested map. Conversely, if $\ell = 0$, monotonicity implies $C_1 = h$. At depth one every seed can be mapped to its tuple $(Y_0, \ldots, Y_d)$ of action outputs without increasing entropy. The finite coupling polytope is compact, so a minimum exists. In a minimizing tuple,

$$H(Y_1, \ldots, Y_d \mid Y_0) = C_1 - H(Y_0) = 0.$$

Thus each Y_i is a function of Y_0, proving the claimed characterization. If any such map is impossible, then $\ell \geq C_1 - h > 0$. This completes the proof of Theorem 1.

7 Relation to the project and scope

The two-action anti-diagonal construction was already present in the project's working note *A Bounded Causal Entropy Tax*, dated 26 September 2026. The extension here replaces its single lower-entropy product law by a seed that simulates an entire adaptive subfamily. The additional technical step is (4): bounding only the entropy of that seed would not justify the absorption argument. The tail-entropy bound also avoids an invalid Fano estimate using the full seed support size.

This closes, at the level of an ordinary mathematical proof relative to the existing finite-seed theorem, the unique-entropy-maximizer question listed in the README. It also gives $L_n = nh + O(1)$ and greedy-seed entropy $nh + O(1)$ by the finite entropy sandwich. It does not identify ℓ in general, resolve the equal-entropy logarithmic problem, or construct one fixed finite-entropy seed for all horizons. Indeed, when $h > 0$, exactness at all horizons forces infinite seed entropy. The finite seeds here may depend on n.

For zero entries, apply the construction on the positive supports and extend the decoder arbitrarily to impossible histories. This gives the same finite-horizon bounds; it is not a new proof of the separate infinite-horizon zero-branch statements.

Verification status. The accompanying Python checks the integer partition construction and exact conditional output probabilities on all histories of small three-action examples. These checks supplement, and do not replace, the proof. No new Lean theorem is supplied or claimed compiled. A novelty claim would require a separate literature assessment.